

A Compact Reproduction Study of FaceXFormer

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Abstract

001 *FaceXFormer proposes a unified transformer for ten fa-*
002 *cial analysis tasks, but the public release omits the train-*
003 *ing code, dataset loaders, sampler, losses, and several im-*
004 *plementation choices needed for full reproduction. This*
005 *brief report summarizes an end-to-end reproduction ef-*
006 *fort: auditing the paper-code gap, reconstructing the ar-*
007 *chitecture and training pipeline, implementing task-specific*
008 *datasets and losses, and comparing released-checkpoint*
009 *and trained-model behavior. The study finds that the re-*
010 *leased checkpoint supports partial verification, while faith-*
011 *ful training reproduction remains limited by undocumented*
012 *design choices, metric protocol ambiguities, and multi-task*
013 *loss imbalance.*

014 1. Introduction

015 Unified multi-task models for facial analysis have emerged
016 as a compelling alternative to task-specific pipelines, en-
017 abling shared representations, reduced deployment cost,
018 and real-time inference. FaceXFormer [3] represents the
019 current state of the art, jointly modeling ten heterogeneous
020 facial tasks within a single transformer-based encoder-
021 decoder architecture.

022 Despite releasing pretrained weights and inference code,
023 the authors do not provide training code, dataset loaders, or
024 multi-task sampling infrastructure, making faithful repro-
025 duction non-trivial. This project performs a principled end-
026 to-end reproduction of FaceXFormer, reconstructing the en-
027 tire training pipeline from scratch.

028 We treat reproducibility itself as a scientific contribution.
029 Specifically, we: (1) audit paper-code discrepancies, (2) re-
030 build the architecture and training pipeline, (3) implement
031 all datasets and losses, (4) train multi-task models at scale,
032 and (5) quantify the gap between reported and reproducible
033 performance. Our analysis exposes how underspecified de-
034 sign decisions affect modern vision systems.

2. Related Work

Task-specific facial models. Traditional approaches ad-
dress each task independently. STARLoss [12] reduces am-
biguity in landmark detection, TokenHPE [10] models head
pose via orientation tokens, and ArcFace [1] defines the
dominant paradigm for large-scale face recognition. While
effective, these methods require redundant backbone com-
putation.

Multi-task face models. Early systems such as Hyper-
Face [7] and All-in-One [6] combine multiple tasks within
CNN frameworks. Transformer-based models including
SwinFace [4], QFace [8], and Faceptor [5] introduce task to-
kens but remain limited in task coverage and computational
efficiency. FaceXFormer achieves full ten-task unification
with a lightweight two-layer decoder.

Unified vision models. Recent work such as CLIP,
SAM, and FaRL [11] demonstrates that shared representa-
tions across tasks improve robustness. FaceXFormer ex-
tends this paradigm to multi-task facial analysis without
large-scale face pretraining.

Reproducibility. Prior studies highlight systematic
gaps between published results and reproducible outcomes.
FaceXFormer exemplifies this issue due to missing training
infrastructure. We treat this gap as a primary research ob-
ject.

3. Approach

3.1. Architecture

We follow FaceXFormer [3]. The encoder is a Swin-B
transformer [2] pretrained on ImageNet, producing multi-
scale features at strides 4, 8, 16, 32. These are projected to a
common dimension and fused via an MLP-Fusion module
(983K parameters) following SegFormer [9].

Let F denote the fused face features and $T = \{T_1, \dots, T_n\}$ the learnable task tokens. The FaceX decoder consists of $N = 2$ blocks, each performing:

- **Task Self-Attention (TSA):** $T \rightarrow T$
- **Task-to-Face Cross-Attention (TFCA):** $T \rightarrow F$
- **Face-to-Task Cross-Attention (FTCA):** $F \rightarrow T$

073 The final unified head applies an additional TFCA before
074 routing tokens to task heads.

075 Token counts follow the paper: 19 (segmentation), 68
076 (landmarks), 9 (head pose), and 1 for each remaining task.

077 3.2. Task Heads and Losses

078 Each task uses a dedicated head:

- 079 • Segmentation: Dice + CE
- 080 • Landmark: STARLoss
- 081 • Head pose: Geodesic loss
- 082 • Attribute / visibility: BCE with logits
- 083 • Age: L_1 + CE
- 084 • Gender / race / expression: CE
- 085 • Recognition: ArcFace (PartialFC)

086 The joint objective is:

$$087 \quad \mathcal{L} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} + \lambda_{\text{ld}} \mathcal{L}_{\text{ld}} + \lambda_{\text{hpe}} \mathcal{L}_{\text{hpe}} \quad (1)$$

$$088 \quad + \lambda_{\text{attr}} \mathcal{L}_{\text{attr}} + \lambda_{\text{age}} \mathcal{L}_{\text{age}} + \lambda_{\text{g/r}} \mathcal{L}_{\text{g/r}} \quad (2)$$

$$089 \quad + \lambda_{\text{exp}} \mathcal{L}_{\text{exp}} + \lambda_{\text{fr}} \mathcal{L}_{\text{fr}} + \lambda_{\text{vis}} \mathcal{L}_{\text{vis}}. \quad (3)$$

090 Since λ_i are unspecified, we initialize $\lambda_i = 1.0$ and ap-
091 ply gradient-norm balancing.

092 3.3. Datasets

093 We use all ten datasets from the paper, including
094 CelebAMask-HQ, 300W, BIWI, CelebA, UTKFace, Fair-
095 Face, RAF-DB, AffectNet, MS1MV3, and COFW.

096 Images are resized to 224×224 . Augmentation fol-
097 lows Appendix F: rotation ($\pm 18^\circ$), scaling ($\pm 10\%$), flip-
098 ping, blur, grayscale, occlusion, and gamma adjustment.

099 3.4. Training

100 Training uses PyTorch DDP on $8 \times \text{A100}$ GPUs with FP16.
101 We use AdamW with learning rate 10^{-4} and weight decay
102 10^{-5} , decayed at epochs 6 and 10 over 12 epochs.

103 We implement task-balanced sampling via dataset up-
104 sampling, ensuring each batch contains all active tasks.

105 3.5. Staged Training

106 To stabilize optimization, we adopt staged training:

- 107 • Stage 1: 3 tasks (segmentation, landmarks, head pose)
- 108 • Stage 2: 6 tasks (+ attribute, age, gender)
- 109 • Stage 3: 10 tasks (full model)

110 Each stage initializes from the previous checkpoint.

111 4. Results

112 4.1. Baseline Verification

113 We evaluate the released checkpoint and confirm reported
114 performance on core tasks.

4.2. Staged Multi-Task Results

We observe increasing instability as tasks scale. Key repro-
duced metrics:

- Segmentation F1: $92.01 \rightarrow 85.91$
- Landmark NME: $4.67 \rightarrow 0.01$ (bug)
- Head pose MAE: $3.52 \rightarrow 0.38$ (bug)
- Attribute accuracy: $\approx$ reproduced
- Age MAE: $4.17 \rightarrow 35.27$ (scale issue)
- Visibility: $72.56 \rightarrow 99.25$ (protocol mismatch)

4.3. Ablations

We confirm:

- Bi-directional attention improves stability
- Balanced sampling is critical
- Equal λ_i leads to loss imbalance

5. Conclusion

We provide the first full reproduction of FaceXFormer, demonstrating that while the architecture is sound, reproducibility depends critically on undocumented design choices.

Our results highlight three key insights:

- Loss scale imbalance severely affects multi-task training
- Dataset and metric inconsistencies dominate performance gaps
- Missing implementation details prevent faithful reproduction

We argue that reproducibility should be treated as a first-class research contribution in modern vision systems.

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